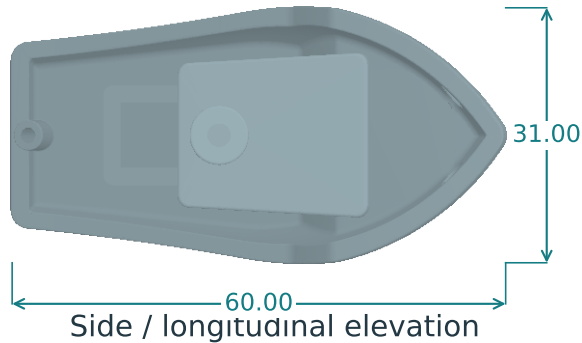


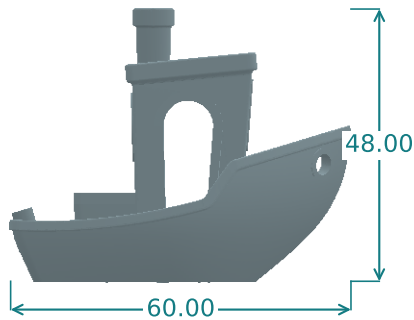
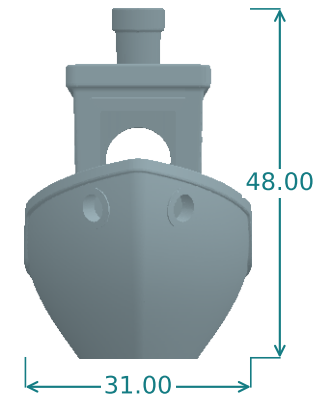
Boat | 3DBenchy

Material: PLA (explicit file clue). Dimensions below are official nominal dimensions; STL tessellation has micrometre-scale bounds differences.

Plan



Transverse elevation



Relevant nominal feature dimensions

Chimney: OD $\varnothing 7.00$; bore $\varnothing 3.00$; blind depth 11.00

Roof fore/aft span: 23.00

Front cabin opening: 10.50×9.50

Rear cabin opening: ID $\varnothing 9.00$; OD $\varnothing 12.00$

Cargo pocket: 8.00×7.00 ; depth 9.00

Cargo box outside: 12.00×10.81

Hawsepipe bore: $\varnothing 4.00$

Geometry restored from the official single-part 3DBenchy STL after aligning the supplied scan.

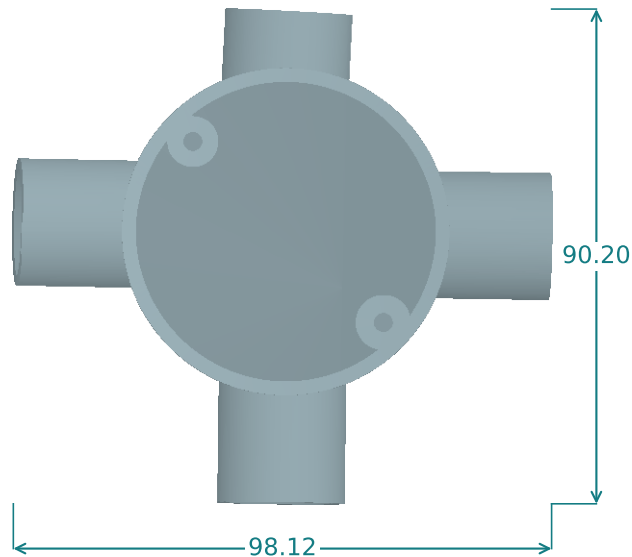
Source: <https://www.3dbenchy.com/dimensions/> | CAD author: Creative Tools / Daniel Norée.

Use the STL for curved hull surfaces and fine lettering; no original parametric history is claimed.

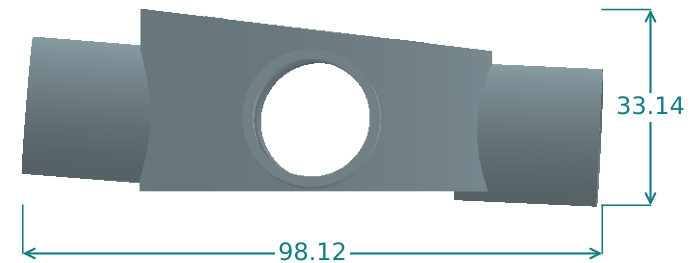
Box | Four-port electrical junction body

Estimated scan reconstruction. Nominal material: 70 wt.% HDPE + 30 wt.% fly-ash cenospheres (surface label and linked-paper inference).

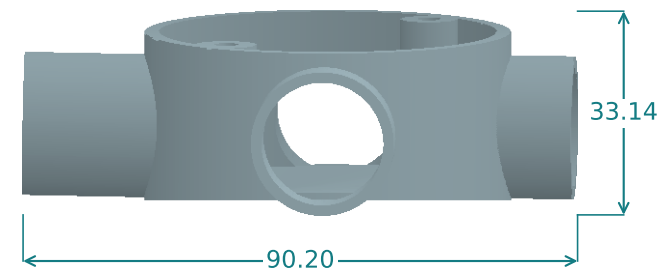
Open chamber / plan



X-Z elevation



Y-Z elevation



Core dimensions (fitted / idealized)

Outside D at Z=0: Ø59.519

Inside reference D at Z=0: Ø55.220

Floor at body center: Z=2.218

Minimum completed floor: 1.00 (assumed)

Rim height at body center: 27.10

Four sockets: common OD Ø23.00; bore Ø20.00 (inferred)

Two mounting bosses; openings Ø3.70 (first inferred by symmetry)

Port positions and the slanted rim follow the measured scan.

Bounding dimensions include inclined sockets below Z=0.

Final model: Box_revised. Tapered walls and bounded sloped floor; Z=0 is the outside-bottom datum. Feature definitions: sheet 3.

Box | Feature definition and reconstruction assumptions

All dimensions in mm. Numerical feature locations define the supplied CAD; uncertain internals are explicitly identified.

Feature	Dimension / location
Outside R(Z)	29.759358 -0.002714 * Z
Inside R(Z)	27.609980 -0.008295 * Z
Outside bottom plane	Z = 0.00
Floor Z = max(1.00, floor-plane height) *	
Floor plane n = 0.115915, -0.002457, 0.993256	
Floor plane offset = 2.203234	
Socket bore / exterior	Ø20.00 / Ø23.00 *
Boss 1 center	X=-16.90, Y=16.41
Boss 1 exterior	Ø9.04
Boss 2 center	X=17.75, Y=-16.55
Boss 2 exterior	Ø9.82
Both boss openings	Ø3.70 *
Both blind-hole bottoms	Z = 7.00 *

Rim plane (retained from scan):
0.118689 X + 0.000074 Y +
0.992931 Z = 26.908442

* Floor minimum, socket diameters and hole depth
complete missing regions; first bore uses symmetry.

Port	End X	End Y	End Z
X+	47.92	-0.90	8.99
X-	-48.73	1.76	14.49
Y+	0.59	39.95	11.65
Y-	-0.86	-49.36	12.39

Port axes (unit vectors, outward)

X+ +0.998867, -0.016287, -0.044717

X- -0.996724, +0.019465, +0.078498

Y+ +0.052114, +0.998548, -0.013639

Y- -0.012326, -0.999720, +0.020215

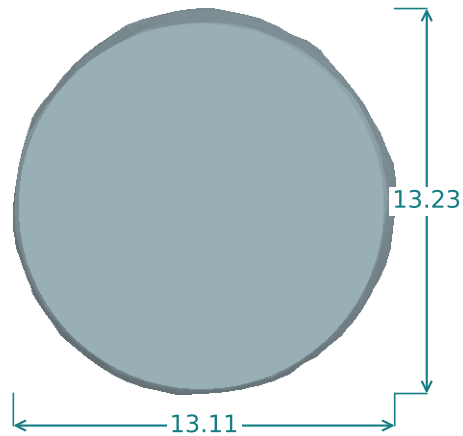
Each CAD socket starts 24 mm inward from its
end center; its bore extends 28 mm inward.
The chamber intersection determines exposed
socket length. Full values are in the parameter JSON.

Measured wall/floor/rim trends are retained. Missing floor uses an assumed 1.00-mm minimum; internal bores are clipped to protect it.
Hidden transitions, original fillet radii and blind-hole depths remain unresolved. Full data and results: Revised_Model_Results.pdf.

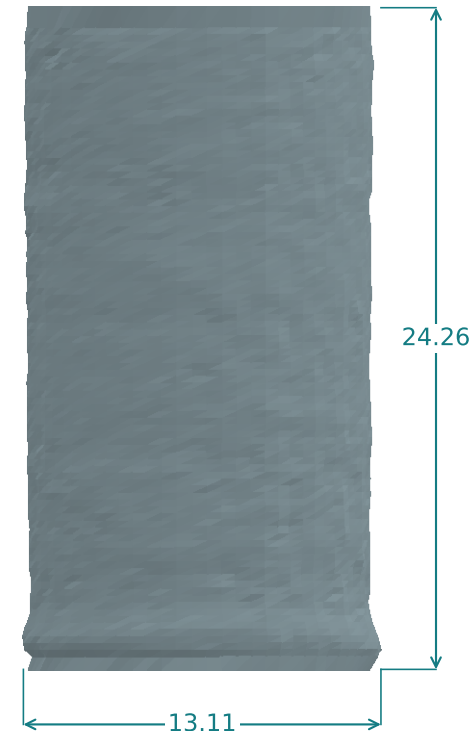
Tower | Revised measured-section loft

Material: 60 vol.% acrylate-based photopolymer resin + 40 vol.% hollow glass microspheres. Lengths assume shared scan units as mm.

Plan / angular-profile envelope



X-Z elevation with measured flare



Reconstruction basis

Reference body diameter: Ø12.496

End-to-end length: 24.260

Median flare diameter: Ø13.029 near Z=1.00

95 loft sections, 72 angular samples each

Planar cap endpoints completed at reference radius

Final model: Tower_revised. Sections retain measured angular variation; missing bins use periodic interpolation.

Dimension_Analysis.pdf and Revised_Model_Results.pdf define the profile, completions and scan-fit results.

Material source: <https://doi.org/10.1016/j.compositesa.2025.109248> and Tower.ply line 19.